

Frobenius Algebras in infinite dimensional Quantum mechanics

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Overview

- Motivation
- Background from Category theory
- Correspondence between orthogonal bases and Frobenius algebras in Finite-dimensional Hilbert Spaces
- Ingredients for infinite-dimensional correspondence
- Some results and the road ahead

Motivation

- No-cloning theorem states that there is no *basis-independent* way to define a (unitary) operator that clones any arbitrary quantum state.
- But starting with an orthonormal basis of a finite-dimensional Hilbert Space, we can *freely* copy these basis elements.
- In finite dimension, as we will see specifying these copy maps *uniquely* determines an orthonormal basis.
- Since Hermitian (more generally Normal) operators have an orthonormal eigen basis, so study of orthonormal basis is important to study of observables in quantum theory.
- Our goal in this talk is to talk about this correspondence and its appropriate extension to arbitrary dimension.

Categories aka boxes and wires

Definition

A category consists of the following data:

- Objects: Labelled by $A, B, C, \dots$
- Morphisms: Which takes an object to another, denoted by $f : A \rightarrow B, g : C \rightarrow D, \dots$

such that if $\text{cod}(f) = \text{dom}(g)$, then there is a morphism $h := g \circ f : \text{dom}(f) \rightarrow \text{cod}(g)$, and this composition is associative and unital.

Categories aka boxes and wires

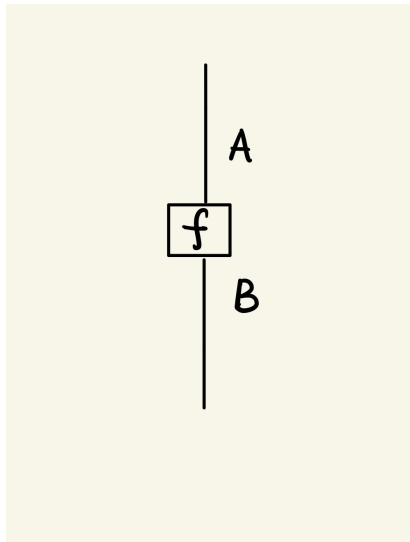


Figure: Boxes and wires

Categories aka boxes and wires

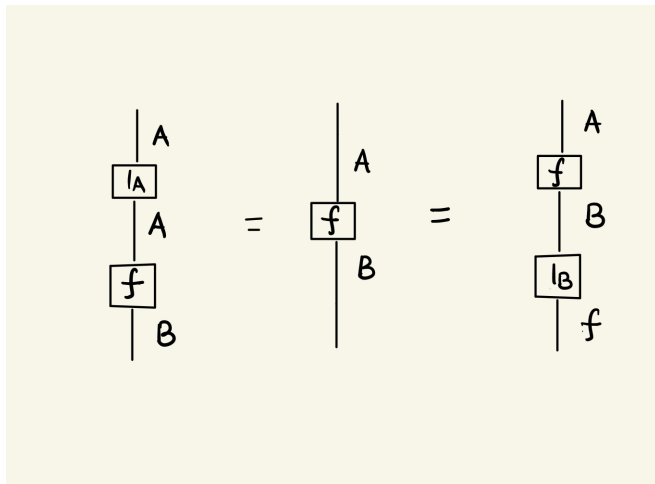


Figure: Unitality of Composition

Examples

- Sets: Objects: Sets, Morphisms: Functions
- Rel: Objects: Sets, Morphisms: Relations
- Grp: Objects: Groups, Morphisms: Group Homomorphisms
- Poset $(A, \leq)$: Objects: Elements of the set A , Morphisms: We define $a \rightarrow b$ iff $a \leq b$.
- FdHilb: Objects: Finite-dimensional Hilbert Spaces, Morphisms: Bounded linear maps.

Maps between Categories

Definition

A functor F from a category $\mathbb{C}$ to a category $\mathbb{D}$ consists of the following data:

- For each object $X \in \mathbb{C}$, there is a $FX \in \mathbb{D}$
- For each morphism $f : X \rightarrow Y \in \mathbb{C}$, there is $Ff : FX \rightarrow FY \in \mathbb{D}$ such that
- $FId_X = Id_{FX}$ for any $X \in \mathbb{C}$
- $F(g \circ f) = (Fg) \circ (Ff)$ for any $f : X \rightarrow Y$ and $g : Y \rightarrow Z \in \mathbb{C}$

Example of Functors

- The free construction of a group on a set is a functorial.
- The forgetful functor $U : Grp \rightarrow Sets$.
- The fundamental Group functor $\pi_1 : Top_* \rightarrow Grp$

Naturality aka no choices involved

Definition

Let $F, G : \mathbb{C} \rightarrow \mathbb{D}$ be two functors, then a natural transformation $n : F \rightarrow G$ is a (family of) maps from $n_X : FX \rightarrow GX$ for each $X \in \mathbb{C}$ such that for any morphism $f : X \rightarrow Y$ in $\mathbb{C}$, the following diagram commutes

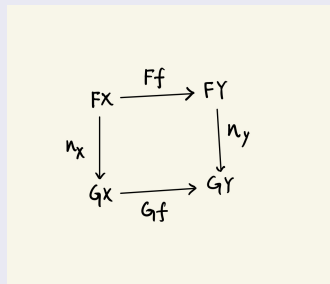


Figure: Naturality

Equivalence aka same information

Definition

An equivalence between categories $\mathbb{C}$ and $\mathbb{D}$ consists of functors $F : \mathbb{C} \rightarrow \mathbb{D}$ and $G : \mathbb{D} \rightarrow \mathbb{C}$ such that $G \circ F \cong 1_{\mathbb{C}}$ and $F \circ G \cong 1_{\mathbb{D}}$

Monoidal Categories aka Parallel composition

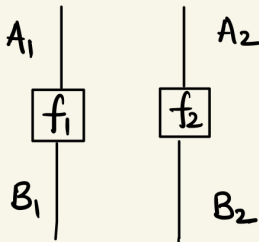
Definition

A (symmetric) monoidal category $\mathbb{C}$ is a category $\mathbb{C}$ equipped with a $\otimes : (A, B) \rightarrow A \otimes B$ and

$\otimes : (f : A \rightarrow B, g : C \rightarrow D) \rightarrow f \otimes g : A \otimes C \rightarrow B \otimes D$ such that:

- Bi-functoriality $\approx$ Between parallel operations, which one you do first doesn't matter
- Symmetric natural isomorphism $\approx$ If you perform the operations and then swap the object or the other way round doesn't matter
- Associativity natural isomorphism $\approx$ The order of tensoring between objects and operations doesn't matter.
- Unitality $\approx$ There exists an empty object I you can discard (or pick it up) in the tensor at will.

Monoidal Categories aka Parallel composition



$$f_1 \otimes f_2: A_1 \otimes A_2 \longrightarrow B_1 \otimes B_2$$

Monoidal Categories aka Parallel composition

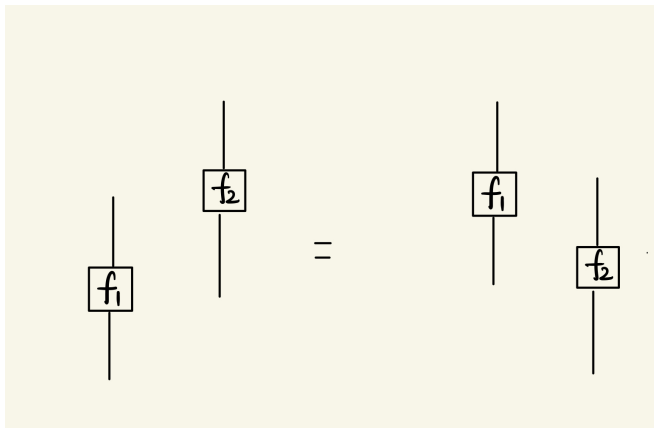


Figure: Which one you do first doesn't matter

Monoidal Categories aka Parallel composition

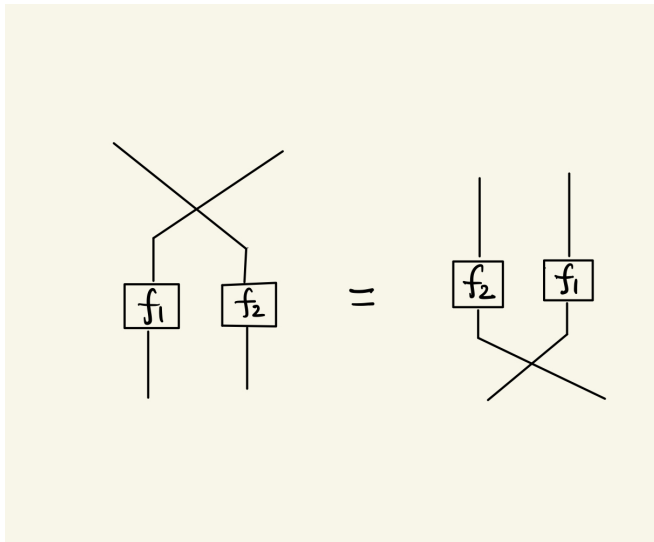


Figure: Swap before or after

Monoid in a monoidal category

Definition

A monoid in a monoidal category $(\mathbb{C}, \otimes, I)$ consists of the following data:

- An object $A \in \mathbb{C}$
- A map $m : A \otimes A \rightarrow A$
- A map $u : I \rightarrow A$

such that the following holds

Monoid in a monoidal category

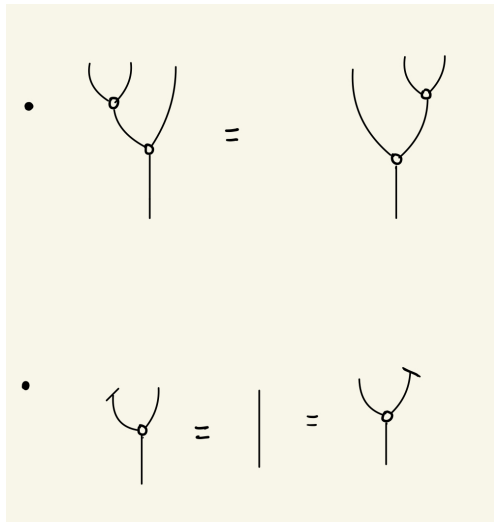


Figure: Monoid laws

Definition

A Frobenius algebra in a monoidal category $(\mathbb{C}, \otimes, I)$ consists of the following data:

- An object $A \in \mathbb{C}$
- An algebra $(A, m : A \otimes A \rightarrow A, u : I \rightarrow A)$
- A co-algebra $(A, \delta : A \rightarrow A \otimes A, \epsilon : A \rightarrow I)$

such that the following holds

Frobenius algebra

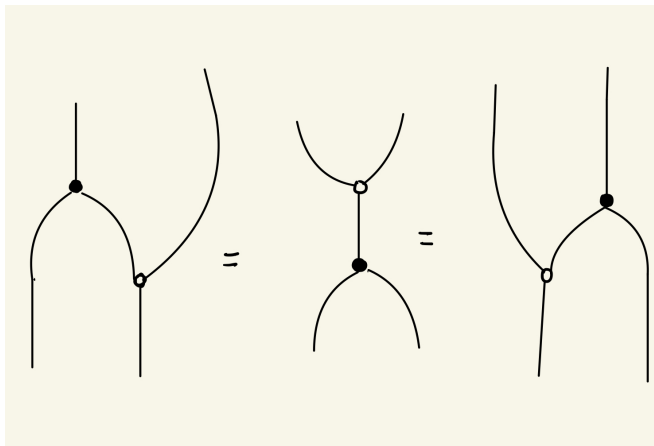


Figure: Frobenius law

From orthogonal basis to Frobenius algebra

- Given a Hilbert space with a chosen orthogonal basis $(H, \{\phi\}_{i=1}^n)$, we define the co-multiplication map by *freely* copying the elements.
- $\delta(\phi_i) = \phi_i \otimes \phi_i$
- $\epsilon(\phi_i) = 1$
- $m := \delta^\dagger, m(\phi_i \otimes \phi_j) = \delta_{i,j} \phi_i$
- $u := \epsilon^\dagger, u(1) = \sum_{i=1}^n \phi_i$

Copyable elements

A copyable element of an algebra on (H, δ, ϵ) is a co-monoid homomorphism $\alpha : I \rightarrow H$

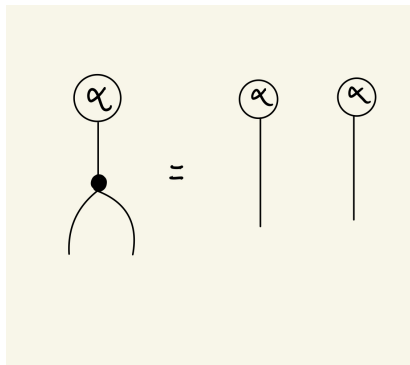


Figure: Copyable elements

From Frobenius algebras to orthogonal basis

- $FdHilb(H, H)$ is a C^* - algebra.
- $(H, m, u) \hookrightarrow (H, H)$ is an involution-preserving embedding, making (H, m, u) a C^* - algebra.
- By structure theorem then $H \cong \bigoplus_{i=1}^n \mathbb{C}$
- By spectral theorem then the adjoint of $1_i := (0, 0, \dots, 1_i, 0, \dots, 0), i \in \{1, 2, \dots, n\}$ will form a basis for H^* , and so $\{1_i\}_{i=1}^n$ will form a basis for H

Inverse constructions

Theorem

These constructions are inverses to each other.

Theorem

The category of $\text{Set}_{\|\cdot\|}$, where the objects are finite sets equipped with a function to the real numbers, and maps are isomorphisms of sets that preserve this norm is equivalent to the category of Commutative Dagger Frobenius algebras on FdHilb with morphisms preserving multiplication and unit.

Theorem

A Frobenius algebra is a monoidal category is an object dual to itself.

Towards infinite dimension

- For any finite dimensional Hilbert Space H , we have $H \cong H^*$
- Categorically, we have maps $n : \mathbb{C} \rightarrow H \otimes H$ and $\epsilon : H \otimes H \rightarrow \mathbb{C}$
- They are defined as follows, given a Hilbert Space H , with an orthonormal basis $\{\phi\}_{i=1}^n$, we define $n := \sum_i |i\rangle \otimes |i\rangle$ and $\epsilon := \sum_i \langle i| \otimes \langle i|$
- For *any* Hilbert space, they have well-behaved duals, in the sense that $H \cong H^{**}$

*-autonomous category

Definition

A $*$ -autonomous category $\mathbb{C}$ is a monoidal closed category equipped with a global dualizing object $\perp$ such that for any object $V \in \mathbb{C}$, $V \rightarrow \mathbb{C}((V, \perp), \perp)$ is an isomorphism.

- For vector spaces, read $\mathbb{C}(V, \perp)$ as V^*
- Given a star-autonomous category we can define a par monoidal structure as $V \oplus W := (V^* \otimes W^*)^*$
- Then $n : \mathbb{C} \rightarrow V^* \oplus V$ and $\epsilon : V \otimes V^* \rightarrow \mathbb{C}$

Definition

Let X be any set and let $\mathcal{U} \subseteq P(X)$, then $\mathcal{U}^\perp := \{S \subseteq X : |S \cap u| < \infty, \forall u \in \mathcal{U}\}$ is called the finitary complement of $\mathcal{U}$. If $\mathcal{U} = \mathcal{U}^{\perp\perp}$, then $(X, \mathcal{U})$ is called a finiteness space.

Morphisms

A morphisms of finiteness spaces $(X, \mathcal{U}), (Y, \mathcal{V})$ is a function (partial function, relations) $f : X \rightarrow Y$ such that it preserves finitary sets and reflects co-finitary sets (sets in $\mathcal{V}^\perp$)

Definition

A Lefschetz Space V is a Topological vector space such that it has an open neighbourhood of open linear subspaces.

Its morphisms are continuous linear maps. The (full) subcategory of Reflexive Lefschetz Spaces is $*$ — autonomous.

Constructing Rlef from Fin-Erhard

- Given a finiteness Space $(X, \mathcal{U})$, we define $\mathbb{C} \langle X, \mathcal{U} \rangle := \{f : X \rightarrow \mathbb{C} : \text{sup}(f) \in \mathcal{U}\}$
- Then $\mathbb{C} \langle X, \mathcal{U} \rangle$ is a vector space with pointwise addition and scalar multiplication.
- We define $V_{u'} := \{f : X \rightarrow \mathbb{C} : \text{sup}(f) \cap u' = \emptyset\}$ and say $S \subseteq \mathbb{C} \langle X, \mathcal{U} \rangle$ is open if for any $s \in S : \exists u' \in \mathcal{U}^\perp : s + V_{u'} \subseteq S$
- Then $(\mathbb{C} \langle X, \mathcal{U} \rangle, \{V_{u'}\}_{u' \in \mathcal{U}^\perp})$ is a Lefschetz Space.

Definition

A linear monoid in $\mathbf{Rlef}$ has the following data:

- A co-monoid $(V, \delta : V \rightarrow V \oplus V, \epsilon : V \rightarrow \mathbb{C})$
- A monoid $(V^*, m : V^* \otimes V^* \rightarrow V^*, u : \mathbb{C} \rightarrow V^*)$
- actions $V^* \otimes V \rightarrow V, V \otimes V^* \rightarrow V$
- co-actions $V^* \rightarrow V \oplus V^*, V \rightarrow V^* \oplus V$
- A morphism of linear monoids is required to preserve the monoid, co-monoid and the dual structure.

Theorem

Erhard's construction is functorial, that is, there is a functor from the category of finiteness spaces (with morphisms as functions or partial functions) to the category of Lefschetz spaces. Importantly, the functor induces basis-to-basis maps (in the case of functions) and basis-to-basis/zero map in $Rlef$.

Interesting theorems

Theorem

In $\mathbb{C} \langle X, \mathcal{U} \rangle$, a series $\sum_{s \in S \subseteq X} f_s$ converges iff $S \in \mathcal{U}$

Theorem

If a series $\sum_{s \in S \subseteq X} \alpha_s f_s$ converges in $\mathbb{C} \langle X, \mathcal{U} \rangle$ for some choice of scalars α_s , then $\sum_{s \in S \subseteq X} \alpha'_s f_s$ converges for any other choice of scalars α'_s . We call this property Scalar Invariant Convergence.

Interesting theorems

Theorem

Any complete Lefschetz Space (with a given basis) has Scalar Invariant Convergence.

Theorem

The Lefschetz spaces constructed from Erhard's construction is always reflexive. It is complete iff $(X, \mathcal{U})$ is a finiteness space. The topology is always Hausdorff and totally disconnected.

Classification theorem

Theorem

Any Complete, Hausdorff, Reflexive Lefschetz space arises as a Finiteness Topology.

Classification theorem

Theorem

Any Complete, Hausdorff, Reflexive Lefschetz space has a topology finer than the product topology and coarser than the discrete topology.

Theorem

Any finite-dimensional Complete, Hausdorff, Reflexive Lefschetz is discrete.

Theorem

There is a functor from the category of Finiteness Spaces (with functions) to the category of linear monoids in $R\text{Lef}$.

Theorem

The copyable elements of a linear monoid in $Rlef$ forms a sub-basis.

Conclusion and the road ahead

- In finite dimensions, choice of an orthonormal basis exactly corresponds to choice of a (commutative dagger) Frobenius algebra on a Hilbert Space.
- In arbitrary dimensions, we must modify our notion of a Frobenius algebra.
- We have shown that there is a functorial construction from the category of Finiteness Spaces to a linear monoid in $\mathbf{Rlef}$, and it passes $\mathbf{Rlef}$, where the induced map is a basis to basis map.
- Coming back, we have shown that the copyable elements does determine a sub-basis for $\mathbf{Rlef}$.
- Looking ahead, the classification theorem we have proven for (Complete, Hausdorff) Reflexive Lefschetz spaces might play a role in proving that the sub-basis determined is actually a basis.

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Thank you!

Questions?